

Glossary of Mathematical Notation

In this appendix, we provide a glossary of mathematical notation not defined in the text.

Notation	Definition and Usage
$\in$	<i>Set membership.</i> $s \in A$ (read “ s is an element of A ” or “ s is in A ”) denotes that the object s (which could be a number, a vector, a set, etc.) is an element of the set A . For example, $2 \in \{1, 2, 4\}$. The negation of $\in$ is denoted by $\notin$; $s \notin A$ (read “ s is not an element of A ” or “ s is not in A ”) denotes that the object s is <i>not</i> an element of the set A . For example, $3 \notin \{1, 2, 4\}$.
$\forall$	<i>For all.</i> Used to state that all elements of a particular set satisfy a given expression. For example, $\forall x \in \{1, 2, 3, 4, 5\}, 2x \leq 10$. The set may be omitted if it is clear from context. For example, $\forall x, x + 1 > x$ might mean that $x + 1 > x$ is true for any real number x . Similarly, $\forall k > 1, f(k) = 0$ might mean that $f(k) = 0$ is true for any integer k that is greater than one. Usually, when defining a function, we will include a “for all” statement after the equation to denote the domain of the function. For example, $g(x) = \sqrt{x}, \forall x \geq 0$.
$\exists$	<i>There exists.</i> Used to state that there is <i>at least one</i> element of a particular set that satisfies a given expression. For example, $\exists x \in \{1, 2, 3, 4, 5\}$ such that $x^2 = 16$. As with $\forall$, the set may be omitted if it is clear from context. For example, $\exists x > 0$ such that $x^2 - 3x - 1 = 0$.

$\iff$	<i>If and only if.</i> Denotes that two statements are logically equivalent; each implies the other. Sometimes written as “iff.” For example, $2x = 10 \iff x = 5$.
$\square$	<i>Quod erat demonstrandum (QED).</i> Latin for “that which was to be demonstrated.” This denotes that a proof is complete.
$\triangle$	<i>End of example.</i> This nonstandard notation allows us to clearly delineate where an example ends.
$\subseteq$	<i>Subset.</i> $A \subseteq B$ (read “ A is a subset of B ”) means that the set B contains every element in the set A . Formally, $A \subseteq B \iff \forall s \in A, s \in B$. For example, $\{2, 3, 5\} \subseteq \{1, 2, 3, 4, 5\}$. Note that $A = B \iff A \subseteq B$ and $B \subseteq A$.
$\emptyset$	The <i>empty set</i> . The set containing no elements. Sometimes written as $\{\}$.
A^C	The <i>complement</i> of a set. When all sets under consideration are subsets of some <i>universal set</i> U , the complement of a set A is the set of all elements in U that are not in A . Formally, $A^C = \{s \in U : s \notin A\}$.
$\cup$	The <i>union</i> of sets. $A \cup B$ (read “ A union B ”) denotes the set containing all elements that are in <i>either</i> A <i>or</i> B (or both). Formally, $A \cup B = \{s : s \in A \text{ or } s \in B\}$. For example, $\{1, 2, 5\} \cup \{2, 3\} = \{1, 2, 3, 5\}$. Union is associative, so for multiple unions parentheses can be omitted without ambiguity. For example, $(A \cup B) \cup C = A \cup (B \cup C) = A \cup B \cup C$.
$\cap$	The <i>intersection</i> of sets. $A \cap B$ (read “ A intersect B ”) denotes the set containing all elements that are in <i>both</i> A <i>and</i> B . Formally, $A \cap B = \{s : s \in A \text{ and } s \in B\}$. For example, $\{1, 2, 5, 6\} \cap \{2, 3, 6\} = \{2, 6\}$. Intersection is associative, so for multiple intersections parentheses can be omitted without ambiguity. For example, $(A \cap B) \cap C = A \cap (B \cap C) = A \cap B \cap C$.
$\setminus$	<i>Set subtraction.</i> $A \setminus B$ (read “ A set-minus B ”) denotes the set that contains all the elements in A <i>except</i> any that are also in B . Formally, $A \setminus B = \{s \in A : s \notin B\}$. $A \setminus B$ is called the <i>relative complement</i> of B in A , since $A \setminus B = A \cup B^C$.
$\mathcal{P}(A)$	The <i>power set</i> of A , that is, the set of all subsets of A . For example, $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}.$

$ A $	The <i>cardinality</i> (or <i>size</i>) of a set. For finite sets, the cardinality is the number of elements in the set. For example, $ \{1, 2, 3, 5, 6, 8\} = 6$.
$\mathbb{N}$	The set of all <i>natural numbers</i> , that is, positive integers: $\mathbb{N} = \{1, 2, 3, \dots\}$. Note that while some texts include zero in $\mathbb{N}$, we do not.
$\mathbb{Z}$	The set of all <i>integers</i> : $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. (The letter “Z” here stands for <i>Zahlen</i> , German for “numbers.”)
$\mathbb{R}$	The set of all <i>real numbers</i> . In classical mathematics, there are many ways of rigorously defining the real numbers, but for our purposes it suffices to say that the real numbers are all the points on a continuous number line.
$\mathbb{R}^n$	The <i>real coordinate space</i> of dimension n , that is, the set of all <i>vectors</i> of length n with real entries. Formally, $\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}, \forall i \in \{1, 2, \dots, n\}\}$. $\mathbb{N}^n$, $\mathbb{Z}^n$, etc. are defined analogously.
$f: S \rightarrow T$	A <i>function</i> f from S to T . The set S is the <i>domain</i> of f , the set of all values s for which $f(s)$ is defined. The set T is the <i>codomain</i> of f , a set that contains all possible values of $f(s)$. Formally, $\forall s \in S, f(s) \in T$.
$f(A)$	The <i>image</i> of the set A under the function f , that is, the set of all values the function f can take on when applied to an element of A . Formally, for a function $f: S \rightarrow T$ and a set $A \subseteq S$, $f(A) = \{t \in T : \exists a \in A \text{ such that } f(a) = t\}$. Note that $f(S)$ is the <i>range</i> of f .
$g \circ f$	The <i>composition</i> of two functions. For functions $f: S \rightarrow T$ and $g: U \rightarrow V$, where $f(S) \subseteq U \subseteq T$, $g \circ f: S \rightarrow V$ is the function that first applies f and then applies g to its output: $\forall s \in S, (g \circ f)(s) = g(f(s))$.
$\sum$	<i>Summation</i> of a sequence. For example, $\sum_{i=1}^5 i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$
$\prod$	<i>Product</i> of a sequence. For example, $\prod_{i=1}^6 (i+1) = (1+1)(2+1)(3+1)(4+1)(5+1)(6+1) = 5,040.$

arg max	The value, among all values in a particular set, that maximizes the given expression. For example, $\operatorname{argmax}_{x \in \mathbb{R}}(-x^2 - 2x + 3) = -1$, as $-x^2 - 2x + 3$ attains its maximum value at $x = -1$. Our use of argmax assumes that there exists a unique maximum. More generally, argmax denotes the set of values at which the maximum is attained.
arg min	The value, among all values in a particular set, that minimizes the given expression. For example, $\operatorname{argmin}_{x \in \mathbb{R}}(x^2 - 4x + 1) = 2$, as $x^2 - 4x + 1$ attains its minimum value at $x = 2$. Our use of argmin assumes that there exists a unique minimum. More generally, argmin refers to the set of values at which the minimum is attained.
$\mathbb{A}^T$	The <i>transpose</i> of a matrix. $\mathbb{A}^T$ denotes the matrix whose columns are the rows of $\mathbb{A}$. Sometimes written as $\mathbb{A}'$. For example, if $\mathbb{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad \text{then} \quad \mathbb{A}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$
$\mathbb{A}^{-1}$	The <i>inverse</i> of a (square) matrix. If $\mathbb{A}$ is <i>invertible</i>, then $\mathbb{A}^{-1}$ denotes the matrix that, when multiplied by $\mathbb{A}$, yields the identity matrix of the appropriate dimensions: $\mathbb{A}\mathbb{A}^{-1} = \mathbb{A}^{-1}\mathbb{A} = \mathbb{I} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$ For example, if $\mathbb{A} = \begin{pmatrix} 2 & 1 & 1 \\ -5 & -3 & 0 \\ 1 & 1 & -1 \end{pmatrix}, \quad \text{then} \quad \mathbb{A}^{-1} = \begin{pmatrix} -3 & -2 & -3 \\ 5 & 3 & 5 \\ 2 & 1 & 1 \end{pmatrix}.$